DeepFusionDTA: Drug-Target Binding Affinity Prediction With Information Fusion and Hybrid Deep-Learning Ensemble Model

Yuqian Pu, Jiawei Li, Jijun Tang, and Fei Guo

Abstract—Identification of drug-target interaction (DTI) is the most important issue in the broad field of drug discovery. Using purely biological experiments to verify drug-target binding profiles takes lots of time and effort, so computational technologies for this task obviously have great benefits in reducing the drug search space. Most of computational methods to predict DTI are proposed to solve a binary classification problem, which ignore the influence of binding strength. Therefore, drug-target binding affinity prediction is still a challenging issue. Currently, lots of studies only extract sequence information that lacks feature-rich representation, but we consider more spatial features in order to merge various data in drug and target spaces. In this study, we propose a two-stage deep neural network ensemble model for detecting drug-target binding affinity, called DeepFusionDTA, via various information analysis modules. First stage is to utilize sequence and structure information to generate fusion feature map of candidate protein and drug pair through various analysis modules based deep learning. Second stage is to apply bagging-based ensemble learning strategy for regression prediction, and we obtain outstanding results by combining the advantages of various algorithms in efficient feature abstraction and regression calculation. Importantly, we evaluate our novel method, DeepFusionDTA, which delivers 1.5 percent CI increase on KIBA dataset and 1.0 percent increase on Davis dataset, by comparing with existing prediction tools, DeepDTA. Furthermore, the ideas we have offered can be applied to in-silico screening of the interaction space, to provide novel DTIs which can be experimentally pursued. The codes and data are available from https://github.com/guofei-tju/DeepFusionDTA.

Index Terms—Drug-target binding affinity, multi-information fusion, multi-channel hybrid neural network, bagging-based ensemble learning

1 Introduction

Identification of novel compounds that interact with the predefined targets is one of the initial steps in drug discovery[1], drug repositioning [2], [3], [4], and drug side effects prediction research[5], [6]. Recently the target profiles and drug effects of massive synthesized compounds are still unidentified. For example, the PubChem dataset contains more than 90 million compounds, but most of them do not have clear interaction profiles. Simultaneously, chemical compound researchers have accumulated large amounts of information on various compound features, properties and target proteins, and constructed numerous large-scale raw data. The purely biological experiment is always accompanied by a lot of time and energy consumption. Therefore, drug researchers desire to use in-silico drug-target interaction prediction methods to process and analyze these high dimensional, complex data, which will help to identity new drugs.

Most of the computational methods are proposed to solve a binary classification problem on drug-target interaction (DTI) prediction [7], [8], [9], [10], [11], [12]. As of last decade, most of the DTI studies utilized the gold standard datasets introduced by *yamanishi et al. 2008* [13] in which pairs of drugs and targets with unknown binding information are treated as negative samples. Recently, DTI studies have been utilizing more realistic datasets created with a chosen binding affinity threshold value [14]. Furthermore, the approximate values of drug-target binding affinity predicted by the regression model help to detect novel drugs with the required interactive properties.

Nowadays, how to indicate the binding affinity between a pair of drug and target is of great significance in the current biology research. Common computational methods are divided into two categories: statistical machine learning model and deep learning model. Among statistical machine learning models, the feature-based method used the known descriptors of the drug and target to generate feature vectors[13], [15], [16], while the similarity-based method based on the premise that similar targets are targeted by similar drugs[17], [18], [19]. KronRLS[20] and SimBoost[21] are two classic statistical machine learning models that predict the binding affinity of each drug-target pair. KronRLS used both different types of drug-drug similarity and protein-

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protein similarity score matrices as input, which was a similarity-based method. SimBoost constructed each drug or target feature based on the similarity networks and described each drug-target pair based on the drug-target network, then established the gradient-enhanced regression tree model to achieve regression tasks.

With the expansion of deep learning application scenarios, deep learning models have been successfully used to solve various problems in bioinformatics applications[22], [23], [24], [25], [26], especially in drug discovery. Compared with statistical machine learning algorithms, deep learning approaches show superior performance in feature extraction of high-throughput data. DeepDTA[27] used label embedding method to encode drug SMILES and protein sequences and constructed a three-layer CNN network to obtain binding affinity value. WideDTA[28] used a text-based method to encode the drug SMILES and protein sequences, including four different textual pieces of information. GANsDTA[29] proposed a novel semi-supervised model based generative adversarial networks(GANs) to predict binding affinity value via drug SMILES and protein sequences.

Although above methods have achieved good performance, accuracy improvement not only depends on the model design, but also the feature richness. For drugs, molecular descriptors are basic computable properties or sophisticated descriptions of a chemical's structure[30]. Concretely, molecular fingerprints are designed to detect the presence of specific compounds in molecular structures, such as circular fingerprint[31], atom environment descriptors (MOLPRINT2D)[32] and molecular access system keys (MACCS)[33]. In addition, a chemical's structural information can also be encoded in text format, such as SMILES[34], which is also the basis for interoperability between various cheminformatics software packages. In recent decades, various machine learning algorithms have been used to predict the drug-target interactions using SMILES or fingerprints as input features [10], [11]. More recently, deep learning models have also been developed in this field [8], [9], [27], [29] .

Structure-based methods that depend on the availability of drug-target complex structures can also predict drug-target binding affinity, but the 3D structure of drugs in complex with a protein is very limited[35], which tends to be a bottleneck for computational speed and accuracy. Therefore, it fails to be applied to predict DTIs on a large scale. Sequence-based methods overcome the limited availability of structural data and the costly need of molecular docking. Rather, they exploit rich omics-scale data of protein sequences, secondary structure and beyond (e.g., biological networks). For proteins, there are many methods that try to predict the secondary structure from protein sequences, such as DeepCNF[36], PORTER 4.0[37], SCORPION[38], SPINE-X[39], SPIDER3 [40], etc.

In this paper, we apply neural network to obtain the merged feature map of candidate drug-target pairs, which is generated by various analysis modules via structure and sequence information, and utilize the ensemble bagging-based lightGBM model to get the approximate binding affinity value, called DeepFusionDTA. For drugs, we use SMILES sequences and Morgan Fingerprint descriptors. For proteins, we utilize amino acid sequences and secondary structure generated by SIPDER3. In the neural network part, we design a Dilated-CNN block constructed by three-layer dilated convolution and a bi-direction long short-term memory (BiLSTM) block since CNNs have advantages in capturing sequence local patterns through weight-sharing strategy and BiLSTM can obtain long-term dependency. We compare different deep neural network designs with various hyper-parameters and information representation. Also, our novel method is compared with the original results of existing outstanding prediction tools.

2 Materials and Methods

In this study, we first design various analysis modules for structure and sequence data, then obtain fusion feature maps of proteins and drugs respectively. Next we reduce the dimensionality of the merged feature maps, which are generated to represent the candidate drug-target pair. Finally, we inject the candidate pair feature map into the ensemble bagging-based lightGBM regression model to get the approximate binding affinity value. The framework of our model is shown in Fig. 1.

2.1 Data Set

We use Davis[41] and KIBA[21], [42] as benchmark datasets to evaluate our proposed model. The Davis dataset contains 442 proteins and 68 drugs, as shown in Table 1. In the Davis dataset, the maximum length of the drug SMILES is 103, the average length is 64, and the maximum length of the protein sequence is 2549, the average length is 788 characters, as shown in Fig. 2. The binding affinity value between drug and target is defined as a logarithmic Kd value, pK_d, as follows:

$$pK_d = -\log_{10}\left(\frac{K_d}{10^9}\right), \tag{1}$$

where the pair with a pK_d value less than 5 is considered to have very weak binding affinity.

The filtered KIBA dataset contains 229 proteins and 2111 drugs, as shown in Table 1. The KIBA binding score optimizes the consistency between K_i, K_d and IC_{50} values. Besides, KIBA guarantees that each protein or each drug contains at least ten interactions. In the KIBA dataset, the maximum length of the drug SMILES is 590, the average length is 58, and the maximum length of the protein sequence is 4128, and the average length is 728 characters, as shown in Fig. 2.

2.2 Input Representation

In this study, structure information and sequence information are used to represent proteins and drugs. In previous studies, the effectiveness of drug fingerprints[8], [43] and protein secondary structure information[44], [45] in predicting drug-target interactions were effectively proved. Rich input information can help the neural network to extract more effective features, which is useful to explore the drug-target interaction mechanism.

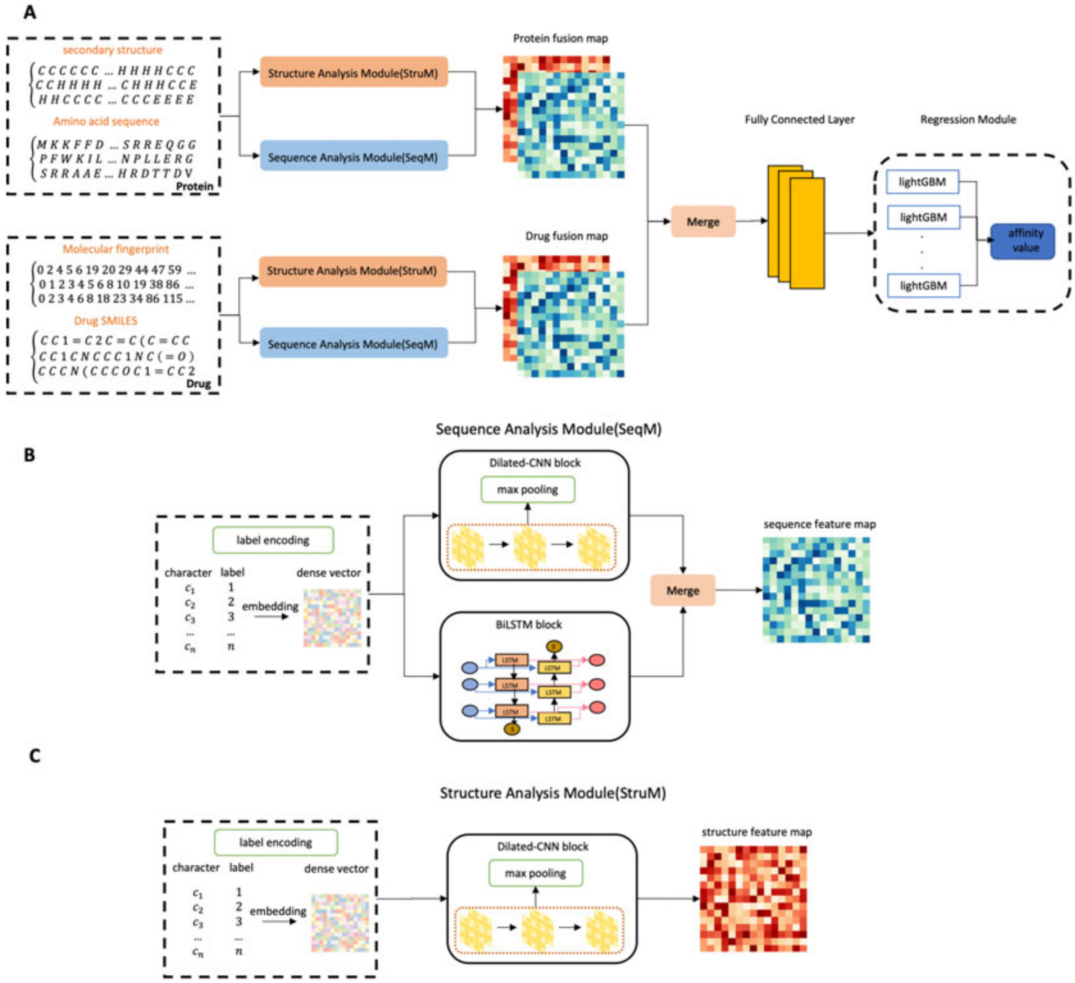

Fig. 1. The flowchart of our proposed method. (A)Input the sequence and structure information of each protein and drug into the sequence analysis module (SeqM) and the structure analysis module (StruM) to obtain protein fusion feature map and drug fusion feature map respectively. Next, concatenate them to generate the candidate drug-target pair feature map by fully-connected layers. Finally, inject the pair feature map into bagging-based lightGBM to predict binding affinity value. (B) The flowchart of the sequence analysis module (SeqM). Sequence information can generate sequence feature map through sequence analysis module, which is composed of Dilated-CNN block and bi-directional long and short-term memory (BiLSTM) block. (C) The flowchart of the structure analysis module (StruM). Considering the importance of local features in structures, we specially apply a single Dilated-CNN block as the structure information analysis module.

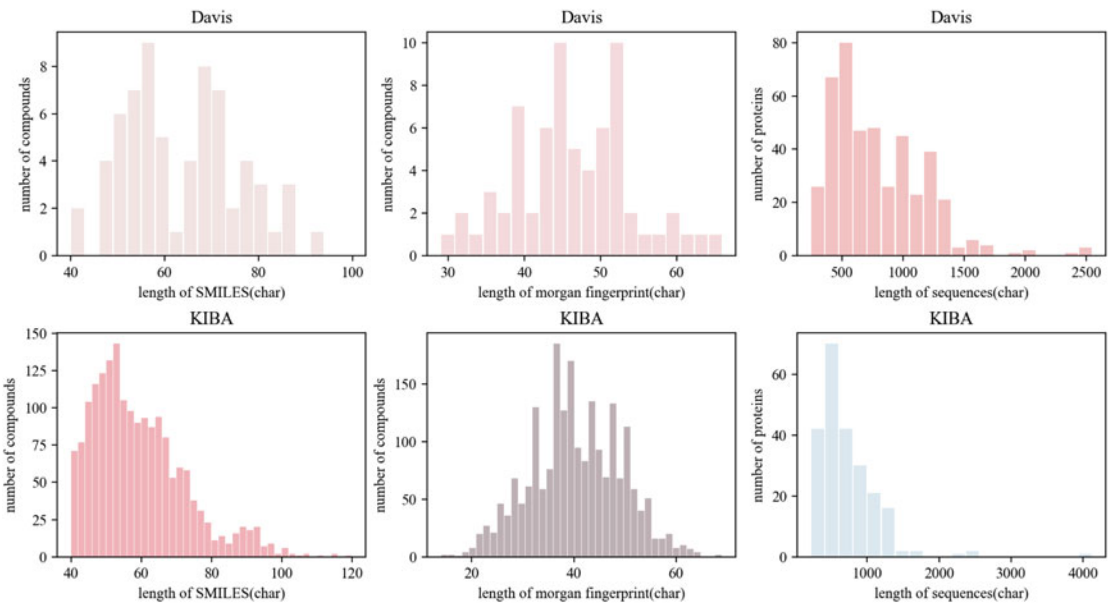

Fig. 2. Data length distribution of Davis and KIBA datasets. The left figure represents the distribution of lengths of SMILES, the middle figure represents the distribution of lengths of morgan fingerprints, and the right figure represents the distribution of lengths of protein sequences.

TABLE 1
Statistics of Davis and KIBA Datasets

	Proteins	Compounds	Interactions
Davis	442	68	30 056
KIBA	229	2,111	118 254

2.2.1 Molecule Representation

Compound SMILES. For compound SMILES, we extract 64 unique letters and utilize label embedding methods to encode the sequences, and a corresponding integer is generated for every unique character.

Molecular Fingerprint. The ability of substructure fingerprints in characterizing drug molecules has been confirmed in some studies[8], [46]. Combining with drug fingerprint information can provide a richer feature description, and it is also convenient for us to study the importance of sequence and structure information in the prediction of drug-target interaction. Given a compound set, morgan fingerprint [31] is applied with a radius of 2 to scan each atom of every compound, and then the corresponding substructure is represented by bit-vector. Morgan fingerprint is generated by RDKit [47]. Hence we utilize the embedding method to convert the substructure information into dense feature vectors to facilitate calculation.

2.2.2 Protein Representation

Protein Sequence. For protein sequences, we extract 25 categories and also utilize label embedding methods to encode the sequences. Here each letter is represented with a unique integer (e.g., ‘C’:1, ‘H’:2, ‘N’:3).

Protein Secondary Structure. Protein secondary structure refers to the spatial arrangement of the backbone atoms of the peptide chain, it also represents a combination of different types of spiral or folded conformations. Protein secondary structure can supply spatial distribution feature in addition to sequence information. Hence, the newly developed software SPIDER3 is used to obtain the secondary structure sequence from protein sequence. For each residue in protein sequence, three possible secondary structure states are predicted: alpha-helix H, beta-strand E, and coil C in SPIDER3. The generated features are considered as the significant resources which can provide local and non-local interaction information of the amino acids. The detailed information about the structure features derived by SPIDER3 are given in [40]. After obtaining the secondary structure representation of the protein by SPIDER3, we apply label embedding method to convert structure information to dense feature matrix.

To ensure the efficiency of coding, both sequences and structures have varying lengths. Based on the distribution of different lengths in different datasets, which can cover at least of 80 percent of the proteins and 90 percent of the drugs as shown in Fig. 2 , we decide on fixed maximum lengths of 85 for SMILES, 60 for morgan fingerprint, 1200 for protein sequence and 1200 for secondary structure for Davis. And we also choose the maximum 100 characters length for SMILES, 60 for morgan fingerprint, 1000 for protein sequence and 1000 for secondary structure for KIBA.

2.3 Neural Network Architecture

In more detail, we design a Dilated-CNN block to capture local effective features via stacked dilated convolution layers, considering that structure and sequence information are both character-based description. In addition, related to possible long-distance dependency in sequence motifs, we also construct a dual-channel sequence analysis module with Dilated-CNN block and bi-direction long short-term memory(BiLSTM) block.

2.3.1 Dilated-CNN Block

Convolution neural network (CNN) is good at using weight-sharing strategy to recognize patterns because the convolution filters can capture more abstract and global features through increasing the number of CNN layers. Patterns are always captured through a series of convolution and pooling operations.

Here, we develop a Dilated-CNN block with three stacked dilated convolution layers[48], which are the improvement of vanilla convolution. For the input area that maps to the filter in the dilated convolution with a dilation constant k, each input unit is away from k units in the vertical and horizontal directions. Dilated convolution can increase the receptive field of the convolution kernel while keeping the number of parameters and the size of the output feature map unchanged.

$$ConV(X)_{i,k} = Leaky\ ReLU\left\{\sum_{m=0}^{M-1} W_m \cdot X_{i\cdot step+k}\right\}, \tag{2}$$

where M represents the length of filter, and k equals to the dilation factor.

We use Leaky ReLU[49] activation function, which is a variation of classic ReLU activation. Leaky ReLU includes a very small slope for negative value inputs, against occurring dead neurons. The equation is as follows:

$$Leaky\ ReLU(x) = \begin{cases} x, \ x \geq 0 \\ \frac{x}{a}, \ x < 0 \end{cases}, \tag{3}$$

where a is a fixed parameter greater than 1.

2.3.2 BiLSTM Block

Another variation of DNNs is the recurrent neural network (RNN). RNN can exhibit dynamic temporal behavior of the internal network state. Here, we build a bi-directional long short-term memory network (BiLSTM), which is a variant of the RNN that combines the outputs of two RNNs, one processing the sequence from left to right, and the other from right to left. The sequence information of drugs and targets are more suitable for BiLSTM, because of good ability of capturing long-term dependency to find contextual relationships. Each LSTM unit is comprised of the input gate, the forget gate and the output gate, the formulation can be expressed as follows:

$$\begin{aligned} f_t &= \sigma(W_f x_t + U_f b_{t-1} + b_f) \\ i_t &= \sigma(W_i x_t + U_i b_{t-1} + b_i) \\ o_t &= \sigma(W_o x_t + U_o b_{t-1} + b_o) \\ C_t &= i_t \circ \tanh(W_c x_t + U_c b_{t-1} + b_c) + f_t \circ C_{t-1} \\ h_t &= o_t \circ \tanh(C_t), \end{aligned} \tag{4}$$

where W and U are parameter matrices and b is a bias vector.

2.3.3 Joint Module

We design a dual-channel sequence analysis module (SeqM) to obtain sequence features via Dilated-CNN block and BiLSTM block that not only can learn the motifs in sequences, but also context dependency. Besides, considering the importance of local features in structures, we specially apply a single Dilated-CNN block as the structure information analysis module(StruM). Fig. 1 shows the structure of sequence analysis module(SeqM) and the structure of structure analysis module(StruM).

Furthermore, for each candidate protein and drug pair, we combine sequence analysis module (SeqM) and structure analysis module (StruM) to obtain feature maps respectively. In the joint module, we concatenate the feature maps as the pair efficient representation, and construct three fully connected layers to reduce the dimensionality of multi-information fusion feature vectors.

For three fully-connected layers with 1024, 1024 and 512 neurons, the output 512-dimension feature vectors are used as new input of the regression module, the equation as follows:

$$FCN(X) = ReLU(W^T X + b), \tag{5}$$

where X is the features vector, W is a $M \times N$ weight matrix and b is a N-dimension bias vector.

2.4 Regression Module

In this study, we achieve better regression performance by strategically combining multiple learning algorithms, which brings two main advantages. On the one hand, in most cases, the true distribution of data cannot be represented by any single hypothesis because of the limited data size. Therefore, weighting multiple hypothetical models can reduce the risk of choosing the wrong model. On the other hand, the ensemble method implements many local search strategies from different starting points, which has advantages compared with a single model such as decision tree or neural network to achieve regression.

Based on the above, we utilize the 512-dimension feature vector, which is output by the fully connected layer, as new input feature of the regression model to perform binding affinity value prediction and adopt the idea of ensemble learning to improve the regression accuracy. We attempt bagging strategy to assemble outputs from multiple lightGBM models that generated by several different subsets of the input features and average for the final prediction. LightGBM has good ability to combine weak and basic learners into a strong learner. We call the lightGBM library with Python to design regression module.

2.5 Model Training and Validation

We randomly divide the dataset into six equal parts in which one part is selected as the independent test set. The remaining parts of the data set are used to complement five-fold cross validation that the four-fold training set for fitting and one-fold for optimizing hyper-parameters. We calculate the mean square error as the loss function, and perform the Adam[50] algorithm to minimize the loss function and optimize the model. Here, we set learning rate as 0.001. Because of the model complexity and the large amount of data, it takes $1-2$ hours for an epoch, and $2-3$ days for each model.

3 Results and Discussion

In this section, we evaluate DeepFusionDTA with the current DTA prediction models. And also, we analyze the impact of different feature representation methods, network architecture, input information selection and regression models on the task of drug-target affinity prediction, in order to understand which factors contribute to the DTA prediction problem.

3.1 Evaluation Criteria

We utilize the Concordance Index (CI) and mean squared error (MSE) metrics to measure the performance of the proposed model. CI score measures whether the predicted affinity values of two random drug-target pairs were predicted in the same order as their true values were, which is shown as below.

$$CI = \frac{1}{Z} \sum_{\delta_x > \delta_y} h(b_x - b_y), \tag{6}$$

where b_x is the prediction value for the larger affinity δ_x, b_y is the prediction value for the larger affinity δ_y, Z is a normalization constant, $h(m)$ is the step function

$$h(m) = \begin{cases} 1, \; if \; m > 0 \\ 0.5, \; if \; m = 0 \\ 0, \; if \; m < 0 \end{cases}. \tag{7}$$

Mean squared error(MSE) measures the difference between the predicted values and labels, the formulation as follows.

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (P_i - Y_i)^2, \tag{8}$$

where P_i is the predicted value, and Y_i is the corresponding actual value.

3.2 Setting of the Hyperparameters

In the feature extraction part, since the input lengths of proteins and drugs are different, grid-search is used to obtain the best filter length settings for proteins and drugs. We select two lengths of 4 and 8 for the drugs and two lengths of 12 and 8 for the proteins to perform grid search. For different parameter combinations, the CI value is shown in the Fig. 3. Three 1D convolution layers are set to 32, 64, and 96 filters respectively. In order to keep the BiLSTM layer

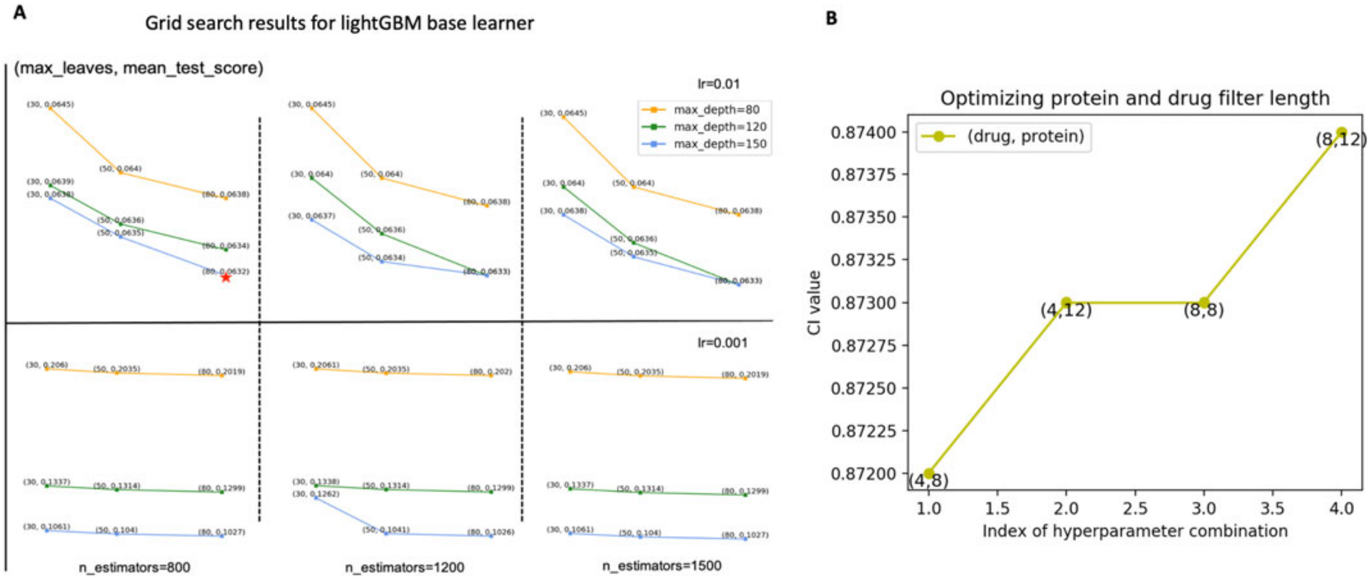

Fig. 3. Neural network and regression model parameters optimization results. (A) Grid optimization of max_depth, n_estimators, max_leaves and learning_rate of lightGBM, the results show that max_depth=150, n_estimators=800, max_leaves=80 and learning_rate=0.01 perform best. (B) Optimizing between the protein filter length [8,12] and the drug filter length [4,8], the results show that 12 for protein filter length, and 4 for the drug filter length can achieve best.

consistent with the output dimension of the convolution layer, 96 hidden nodes are also used to concatenate the output of the convolution layer through an averaging operation. See the Table 2 for details about the settings of optimized parameters in the neural network.

In the bagging-based lightGBM module part, we set the learning rate, the number of leaves, the number of estimators and the max depth as hyperparameters. We also apply grid search to explore the best parameters settings of lightGBM base learner as Fig. 3, gradient boosting decision tree (GBDT) mode is chosen, the learning rate is set to 0.01, the number of leaves is 80, and 800 base learners are constructed during the model training. The max depth is fixed to 150, which can control over-fitting. In bagging learning, we adopt 800 estimators to accomplish regression.

3.3 Verify Effectiveness of Structure Information

Feature selection is another important step, where the selection will directly affect the performance of feature extraction. In order to study the contribution of structure information in solving DTA problems, we also employ the prediction performance of the proposed model with and without structure information. Input with structure information can achieve the best result with 0.876 CI value and 0.176 MSE value and input without structure can achieve the best result with 0.868 CI value and 0.187 MSE value. There is a **0.008** improvement in CI and a **0.011** improvement in MSE. The results reveal that structure information plays an essential role in DTA prediction. Compound morgan fingerprint and protein secondary structure are obtained by SMILES and amino acid sequences respectively, which contain unique identification, ultimately determines the spatial structure and feature of drugs and proteins.

TABLE 2
Parameter Settings for Deep Learning Module

Hyperparameters	Value
Number of filters	32;64;96
Filter length(drugs)	[8,4]
Filter length(proteins)	[12,8]
epoch	100
number of BLSTM uints	96
FCN hidden neurons	1024;1024;512
batch size	256
optimizer	Adam
learning rate(lr)	0.001

3.4 Performance of Various Input Encoding Methods

In order to improve the precision of the DTA prediction, it is very important to choose an effective input encoding method to describe the effective feature of input. Therefore, we use one-hot encoding, label embedding and word2vec [51] methods to study the effectiveness of encoding method. One-hot encoding means using 1 in different positions to represent the unique letter. Word2vec has achieved excellent performance in solving natural language processing (NLP) problems[52], [53]. Based on the fact that words in similar environments usually have similar meanings, the word2vec model predicts the central word based on adjacent words in a given window. [54] utilized word2vec method to study the drug-target interaction prediction on different dataset and we want to study whether this method is proper to our datasets. The best parameters of word2vec are determined by grid search.

Take the KIBA dataset as an example, it is obvious to see that the representation is more accurate when the label embedding method is used to encoding the input, where the MSE value is 0.176 and CI value is 0.876 as shown in the Table 3. Comparing between one-hot and word2vec, the performance of label embedding is better. The one-hot coding matrix is too sparse, and whether word2vec can

TABLE 3
Study the Impact of Different Factors on the Accuracy of the Model on the KIBA Dataset

model	input encoding			architecture			regression model				ensemble learning		CI	MSE
	one-hot	word2vec	label embedding	CNN	CNN+BiLSTM[1]	CNN//BiLSTM[2]	XGBoost	RF	ridge	lightGBM	stacking	bagging		
	✓					✓				✓		✓	0.874	0.175
		✓				✓				✓		✓	0.812	0.256
			✓			✓				✓		✓	0.868	0.187
			✓	✓						✓		✓	0.863	0.194
			✓		✓					✓		✓	0.869	0.181
			✓			✓	✓					✓	0.868	0.778
			✓			✓		✓				✓	0.873	0.173
			✓			✓			✓			✓	0.871	0.173
			✓			✓	✓	✓	✓	✓	✓		0.871	0.177
DeepFusionDTA			✓			✓				✓		✓	**0.876**	0.176

[1]*CNN+BiLSTM represents CNN followed with BiLSTM.*
[2]*CNN//BiLSTM represents dual-channel built-in with CNN and BiLSTM.*

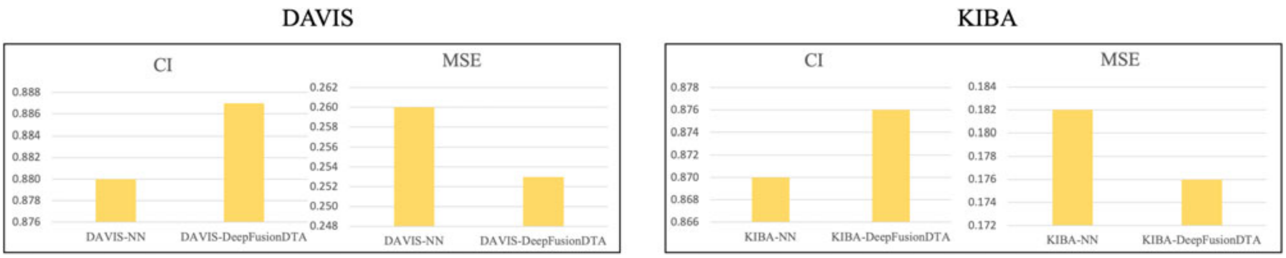

Fig. 4. Comparison of the performance of Davis and KIBA datasets on whether to use ensemble learning strategy. *-NN means the neural network not only extracts the features, but also obtains the predicted value through the linear layer, *-DeepFusionDTA means the neural network extracts the features and applies the ensemble learning strategy to obtain the predicted value.

accurately express the similarity of proteins without prior knowledge remains to be studied, and it may not be suitable for our dataset.

3.5 Performance of Various Network Architectures

CNN captures the pattern of sequence and structure information and BiLSTM is capable of finding the long-term dependency, but different network design methods also have essential impact on extracting effective information. There are three common types of frameworks, including only CNN, CNN followed with BiLSTM and dual-channel built-in with CNN and BiLSTM. Different types of architecture are tested for performance comparison, the results on the KIBA dataset are shown in Table 3.

The structure of stacked CNN obtains CI value 0.863 and MSE value 0.194, CNN followed with BiLSTM obtains CI value 0.869 and MSE value 0.181, dual-channel built-in with CNN and BiLSTM obtains CI value 0.876 and MSE value 0.176. The results indicate that dual-channel built-in with CNN and BiLSTM achieves the best performance on the two metrics. Dual-channel feature extraction could be better to learn long-distance dependency and pattern information independently, and the fusion feature generated by concatenating the dual-channel outputs can effectively improve the accuracy.

3.6 Performance of Various Ensemble Learning Methods and Regression Models

In order to construct an effective DTA prediction model, the choice of regression model is crucial. To evaluate the performance of lightGBM, we compare lightGBM with XGBoost, Ridge regression and Random Forest (RF) on the KIBA dataset. Through grid search, we set the optimal parameters for above models. The performance on KIBA dataset is shown in the Table 3. The XGBoost model has 0.187 MSE value and 0.868 CI value; the Ridge has 0.177 MSE value and 0.871 CI value; the RF model has 0.173 MSE value and 0.874 CI value, and the lightGBM model has 0.176 MSE value and 0.876 CI value on the KIBA dataset, which is more suitable for our regression task.

There are two common types of ensemble learning, including bagging and stacking. The idea behind stacking is to use the predicted labels of base learners as new input features for the meta-learner. The meta-learner is able to reduce the errors made by each of the base learner through its training process, which can finally prove the generalization ability of the system. In order to study which ensemble learning strategy is more effective, we construct a stacking module that choose Ridge, RF and XGBoost as base learner, and lightGBM as meta-learner. The performance on KIBA dataset is shown in the Table 3.

In addition, we also verify the effectiveness of setting the regression module and prove that it can improve the regression accuracy based on the feature extraction by deep learning.The performance on Davis and KIBA datasets are shown in the Fig. 4. We find that the neural networks can extract deep semantic features. Simultaneously, ensemble learning can more effectively use precise features to build regression models.

3.7 Comparison to Other Outstanding Methods

In order to evaluate the performance of our proposed method, DeepFusionDTA, we also compare with three other methods, namely DeepDTA, WideDTA and GANsDTA on

TABLE 4
The Average CI and MSE Scores of the Test Set for the Davis Dataset

Models	target rep.	drug rep.	CI	MSE	p-value
KronRLS	S-W	Pubchem-Sim	0.871	0.379	
SimBoost	S-W	Pubchem-Sim	0.872	0.282	
DeepDTA	S-W	Pubchem-Sim	0.790	0.608	
DeepDTA	S-W	CNN	0.886	0.420	
DeepDTA	CNN	Pubchem-Sim	0.835	0.419	
DeepDTA	CNN	CNN	0.878	0.261	6.58e-67
GANsDTA	GAN	GAN	0.881	0.276	
WideDTA	PS&PDM	LS&LMCS	0.886	0.262	5.64e-65
DeepFusionDTA	SeqM&StruM	SeqM&StruM	**0.887**	**0.253**	

TABLE 5
The Average CI and MSE Scores of the Test Set for the KIBA Dataset

Models	Target rep.	Drug rep.	CI	MSE	p-value
KronRLS	S-W	Pubchem-Sim	0.782	0.411	
SimBoost	S-W	Pubchem-Sim	0.836	0.222	
DeepDTA	S-W	Pubchem-Sim	0.710	0.502	
DeepDTA	S-W	CNN	0.854	0.204	
DeepDTA	CNN	Pubchem-Sim	0.718	0.571	
DeepDTA	CNN	CNN	0.863	0.194	6.99e-04
GANsDTA	GAN	GAN	0.866	0.224	
WideDTA	PS&PDM	LS&LMCS	0.875	0.179	9.13e-06
DeepFusionDTA	SeqM&StruM	SeqM&StruM	**0.876**	**0.176**	

the Davis dataset and KIBA dataset. As shown in Tables 4 and 5, DeepFusionDTA achieves best performance with 0.887 CI and 0.253 MSE value on the Davis dataset and 0.876 CI, 0.253 MSE on the KIBA dataset. However, DeepDTA performs well with 0.878 CI and 0.261 MSE value on the Davis dataset and 0.863 CI, 0.194 MSE on the KIBA dataset; WideDTA performs well with 0.886 CI and 0.262 MSE value on the Davis dataset and 0.875 CI, 0.179 MSE on the KIBA dataset; GANsDTA performs well with 0.881 CI and 0.276 MSE value on the Davis dataset and 0.866 CI, 0.224 MSE on the KIBA dataset. According to the Pearson coefficient metric provided by WideDTA, the Pearson coefficient of DAVIS dataset is 0.820, and the Pearson coefficient of KIBA dataset is 0.856. For the results of our model, the Pearson coefficient of DAVIS dataset is 0.830, and the Pearson coefficient of KIBA dataset is 0.865. Compared with the CI and MSE metrics, the Pearson coefficient has been improved. In addition, we have also calculated the p-value of WideDTA and DeepDTA using available resources.The results show that our model is significantly different from these two works. Overall, our model can obtain a good result. And this proves that various analysis modules for structure and sequence and bagging-based lightGBM regression model incorporates some unique features to outperform other methods for the drug-target binding affinity prediction.

4 Conclusion

We propose a machine-learning based approach to predict drug-target binding affinity using sequences and structure information of proteins and drugs. We first design a dual-channel sequence analysis module to obtain sequence features via Dilated-CNN block and BiLSTM block, meanwhile specially apply a single Dilated-CNN block as the structure information analysis module. Hence, we inject the candidate pair feature map that is generated by 512-dimension fully connected layer into the ensemble bagging-based lightGBM regression model to get the approximate binding affinity value. By designing different feature extraction modules for structure and sequence and combining bagging-based lightGBM model, we are able to achieve better performance than the current methods, and we also employ different impacts on feature selection and framework to compare the performance of our proposed model.

The finding of our study suggests that only sequences feature and stacked CNN layers are not sufficient to describe drug-target interaction mechanism. When utilizing sequence and structure information as the fusion feature and various analysis modules, the performance can get better improvement compared with other models on both KIBA and Davis datasets.

It is plausible that a number of limitations may have influenced the results obtained. The most crucial is that the model using the structure information represented by sequence is weaker than other represented by topology. This might be an indication that the atoms of drugs and amino acids require a connection map that can handle their ordered relationships, which the vanilla neural network architecture failed to capture successfully. Graph Neural Network (GNN), which is a special type of neural network based graph representation, could be a more suitable approach to learn from topology of drugs and targets[55], [56], [57]. A large percentage of drug-target pairs remain unknown interactions, as future work, we will focus on building an effective representation for protein and drugs topology.

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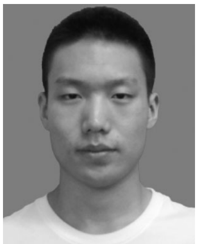

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